



```
#include<stdio.h>
int main()
{
    int n,r,i,j,k;
    printf("Enter the number of processes:");
    scanf("%d",&n);
    printf("Enter the number of resource types:");
    scanf("%d",&r);
    int alloc[n][r],max[n][r],need[n][r],avail[r];
    int finish[n],safe_sequence[n];
    int count=0,safeFound;
    printf("Enter the allocation Matrix:\n");
        for(i=0;i<n;i++){
            for(j=0;j<r;j++)
            {
                scanf("%d",&alloc[i][j]);
            }
        }
    printf("Enter Max matrix:\n");
    for(i=0;i<n;i++)
    {
        for(j=0;j<r;j++)
        {
```

```
for(i=0;i<n;i++)
{
    for(j=0;j<r;j++)
    {
        scanf("%d",&max[i][j]);
    }
}
printf("Enter the Available Resources:\n");
for(i=0;i<r;i++)
{
    scanf("%d",&avail[i]);
}
for(i=0;i<n;i++)
{
    finish[i]=0;
}
for(i=0;i<n;i++)
{
    for(j=0;j<r;j++)
    {
        need[i][j]=max[i][j]-alloc[i][j];
    }
}
```

```
        need[i][j]=max[i][j]-alloc[i][j];
    }
}
while(count<n)
{
    safeFound=0;
    for(i=0;i<n;i++)
    {
        if(finish[i]==0)
        {
            int safe=1;
            for(j=0;j<r;j++)
            {
                if(need[i][j]>avail[j])
                {
                    safe=0;
                    break;
                }
            }
            if(safe)
            {
                for(k=0;k<r;k++)
                {
```

```

        if(safe)
        {
            for(k=0;k<r;k++)
            {
                avail[k]=avail[k]+alloc[i][k];
            }
            safe_sequence[count]=i;
            finish[i]=1;
            safeFound=1;
            count++;
        }
    }
    if(!safeFound)
    {
        printf("System is in unsafe state!May occur the deadlock
\n");
        return 0;
    }
}
printf("Safe sequence is:");
for(i=0;i<n;i++)
{

```



```
                                avail[k]=avail[k]+alloc[i][k];
                                }
                                safe_sequence[count]=i;
                                finish[i]=1;
                                safeFound=1;
                                count++;
                                }
                                }
                                }
                                if(!safeFound)
                                {
                                    printf("System is in unsafe state!May occur the deadlock
\n");
                                    return 0;
                                }
                                }
                                printf("Safe sequence is:");
                                for(i=0;i<n;i++)
                                {
                                    printf("P%d ",safe_sequence[i]);
                                }
                                return 0;
                                }
```

```
Enter the number of processes:5
Enter the number of resource types:4
Enter the allocation Matrix:
2 0 0 1
3 1 2 1
2 1 0 3
1 3 1 2
1 4 3 2
Enter Max matrix:
4 2 1 2
5 2 5 2
2 3 1 6
1 4 2 4
3 6 6 5
Enter the Available Resouces:
3 3 2 1
Safe sequence is:P0 P3 P4 P1 P2 [varsha@localhost cpuscheduling]$
```